

# Tenderly

AI-Powered Volunteer Matching for San Francisco

**When someone has the willingness to help,  
Tenderly turns uncertainty into a clear next step.**

AI for Social Good — MLH & DigitalOcean Hackathon · July 2026

# The Problem

People want to help but hit three kinds of friction:

- 1 They don't know which organization needs *their* particular skills
- 2 Long opportunity lists feel overwhelming
- 3 Urgency is invisible — volunteers can't tell when help matters most

Meanwhile, nonprofits compete for attention even as conditions change across the city.

## The Gap

50 listings to browse

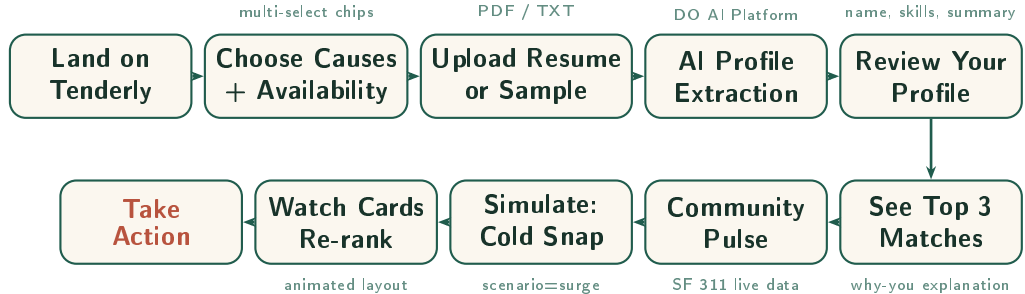


No clear next step

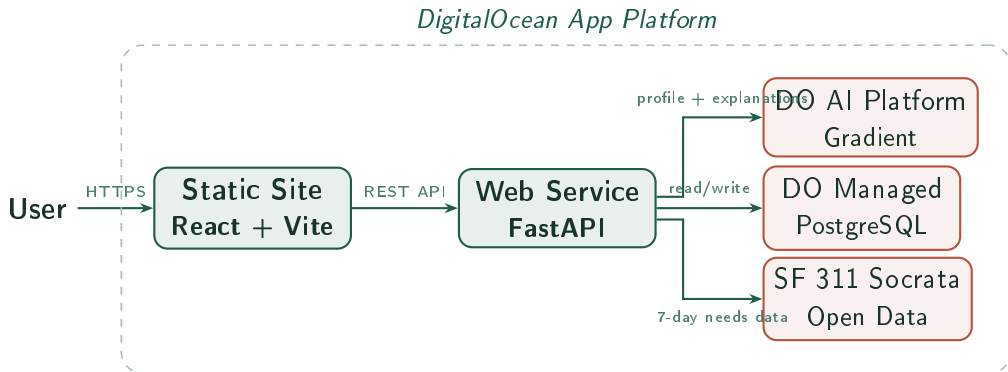


Volunteer gives up

# User Journey



# DigitalOcean Platform Usage



## Static Site

Frontend built from Vite

## Web Service

FastAPI backend on port 8080

## AI Platform

2 calls per user journey

# How Matching Works

## Weighted Match Score

$$\text{score} = \underbrace{0.45}_{\text{skills}} + \underbrace{0.25}_{\text{causes}} + \underbrace{0.20}_{\text{avail.}} + \underbrace{0.10}_{\text{location}}$$

**Skills** — do your abilities match the org's needs?

**Causes** — do you care about the same issues?

**Availability** — can you show up when needed?

**Location** — how active is need nearby?

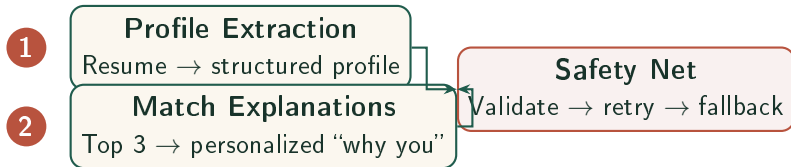
## What Happens in a Cold Snap?

- Food & shelter roles get a **score boost**
- Higher-need neighborhoods rise in ranking
- Urgency labels update (medium → high)

## Explainable, Not a Black Box

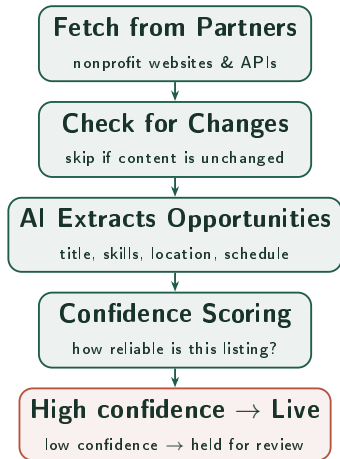
Every match includes a plain-language reason so the volunteer knows *why*.

# DigitalOcean AI Platform — Two Calls Per Journey



**Key guarantee:** If the AI service is down, responses fall back to pre-written defaults. The app never breaks.

# How Opportunities Get Into Tenderly



## Runs Every 6 Hours

A scheduled job pulls the latest listings from partner nonprofits automatically.

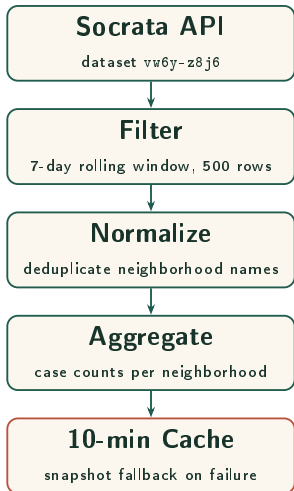
## Smart Deduplication

Each import is fingerprinted. Unchanged pages are skipped. Changed parsing rules trigger re-evaluation.

## Source Isolation

If one partner's site goes down, all other sources still import normally.

# Community Needs: Live SF 311 Data



## Data Rules

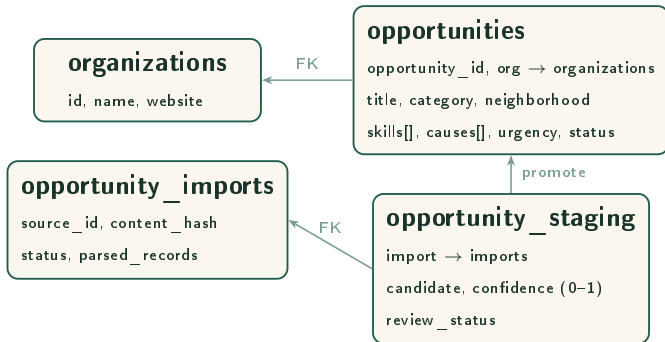
- Freshness label with timestamp
- Stale data labelled honestly
- Counts as context, not crisis claims
- Top 8 neighborhoods shown

## Fallback Guarantee

If Socrata is unreachable, a bundled snapshot is served with the same format — the UI never breaks.



# Data Model



Every recommendation traces back to an approved source and parser run.

# Tenderly

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Deployed on DigitalOcean App Platform

Powered by DigitalOcean AI Platform · Live SF 311 Open Data · PostgreSQL